



# Dr. WAQAS AHMAD

Researcher

+971 54 544 0961  
waqasahmad000713@gmail.com  
orcid.org/0000-0003-2160-3962  
Sharjah, United Arab Emirates

## SUMMARY

As an accomplished engineering academic, I hold a PhD in Materials Science and Engineering and have gained several years of experience in university teaching, laboratory instruction, and applied research in both the UAE and Pakistan. My work focuses on sustainable energy systems, thermochemical conversion of solid wastes, waste-to-energy, biochar-based resource recovery, and low-carbon technologies, supported by peer-reviewed publications and industry-linked research. I have experience teaching and mentoring undergraduate and postgraduate students, supporting laboratory- and project-based learning, supervising student projects, and contributing to curriculum delivery and academic service. Committed to excellence in teaching, impactful research, and student success, I am well prepared to contribute to an assistant/associate professor role through interdisciplinary collaboration, innovation in engineering education, and sustainability-focused scholarship.

## RESEARCH & ACADEMIC EXPERIENCE

### Assistant Professor - Mechanical Engineering

02/2026 - Present

Amity University Dubai, Dubai, UAE

- Teach undergraduate courses including Thermodynamics, Fluid Mechanics, Basic Mechanics, Engineering Materials, Engineering Graphics and Workshop Practices.
- Conduct and supervise laboratory sessions for Workshop Practices Lab, Basic Mechanics Lab, and Fluid Mechanics Lab.
- Participate in examination committee meetings and academic review processes.
- Contribute to CEO board meetings focused on Master's curriculum development and academic program enhancement.

### Teaching Assistant - Environmental Sciences

01/2022 - 01/2026

American University of Sharjah, Sharjah, UAE

- Supported undergraduate instruction under Prof. Fatin Samara in advanced analytical techniques (SEM/EDS, ICP-AES, BET, GC-MS), delivered focused lectures on the principles, operation, and applications of analytical instruments, and assisted students with data interpretation.
- Served as a Teaching Assistant for CHM345 "Instrumental Analysis" under Prof. Mustafa Khamis, provided approximately 4 hours per week of in-lab instruction, trained students in analytical instrument operation and sample preparation, guided data interpretation, and prepared and delivered final result reports to students.

### Research Assistant - Materials & Environmental Science

08/2021 - 01/2026

American University of Sharjah, Sharjah, UAE

- Worked on an industry-funded Ph.D. project with Emirates Global Aluminium on acidic biochar production through pyrolysis for soil amendment and waste valorization (industrial project value: ~1.6 million AED).
- Operated batch tube and auger reactors to produce biochar, bio-oils, and biogas, ensuring proper system operation and thorough documentation of process conditions and product yields.
- Conducted toxicity assessments, including heavy metal analysis using SEM/EDS and ICP-AES; evaluated aromatic hydrocarbons and dioxins/furans using FTIR and GC/MS; and assessed biochar surface behavior across varying pH conditions using zeta potential analysis with the Litesizer 500.
- Performed additional academic responsibilities, including grading examinations, delivering lectures, assisting students with experimental procedures and data analysis, and guiding students in research activities, scientific writing, and thesis preparation.
- Received formal training on advanced analytical instruments such as ICP-AES, AAS, and GC, and subsequently developed standard operating procedures (SOPs) for ICP-AES and the Litesizer 500 following certification.

### Lecturer in Mechanical Engineering

01/2020 - 08/2020

National University of Computer and Emerging Sciences (FAST-NUCES), Islamabad, Pakistan

- Taught undergraduate courses in Basic Mechanical Engineering and related subjects, supporting students in foundational mechanical concepts.
- Maintained the iLearn learning management system, tracked Course Learning Outcomes (CLOs) and Program Learning Outcomes (PLOs), and prepared documentation for Quality Enhancement Cell (QEC).

### Research Assistant - Waste-to-Energy & Biogas

02/2018 - 04/2019

National University of Sciences & Technology (NUST), Islamabad, Pakistan

- Worked on a USAID-funded project: Design and fabrication of semi-continuous stirred tank reactors for anaerobic digestion of poultry manure.
- Assembled and operated anaerobic digestion reactors, including daily feeding and monitoring for stable operation.
- Performed gas chromatography on biogas samples to characterize composition and evaluate process performance.

### Project Consultant - Education & Development

08/2017 - 02/2018

Iqra Fund, Islamabad, Pakistan

- Contributed to a USAID-funded project: Providing equitable, quality and sustained school education and vocational training to vulnerable communities (budget ~7.9 million PKR).
- Prepared comprehensive project management plans, including scope, schedule, cost, and risk management components.

## INDUSTRIAL INTERNSHIPS (EARLY CAREER)

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### Suzuki Motors, Peshawar, Pakistan

08/2016 - 10/2016

Assisted with vehicle maintenance and workshop documentation.

### British American Tobacco Company, Akora Khattak, Pakistan

07/2015 - 09/2015

Participated in 5S implementation in the maintenance workshop; supported maintenance tasks.

### Hydel Power Station, Tarbela Dam, Topi, Pakistan

06/2014 - 07/2014

Observed and assisted in power plant maintenance and operations.

## EDUCATION

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### PhD in Materials Science and Engineering

08/2021 - 11/2025

American University of Sharjah (AUS), Sharjah, United Arab Emirates

- GPA: 3.46 / 4.00
- Thesis: Potential of Co-Feeding Pyrolysis in Arid Regions for Energy Production, Carbon Sequestration, and Soil Amendment

### MSc in Thermal Energy Engineering

08/2017 - 01/2020

U.S.-Pakistan Center for Advanced Studies in Energy (USPCAS-E), NUST, Islamabad, Pakistan

- GPA: 3.55 / 4.00
- Thesis: Feasibility and Experimental Validation and Cost-Benefit Analysis of Municipal Solid Waste to Energy

### MSc in Project Management

12/2016 - 01/2018

Shaheed Zulfikar Ali Bhutto Institute of Science and Technology (SZABIST), Islamabad, Pakistan

- GPA: 3.41 / 4.00
- Thesis: Providing equitable, quality, and sustained school education and vocational training to vulnerable communities

### BSc in Mechanical Engineering

08/2012 - 05/2016

University of Engineering and Technology (UET), Peshawar, Pakistan

- GPA: 3.13 / 4.00
- Thesis: Testing and making of refuse-derived fuels

## HONORS, AWARDS & SCHOLARSHIPS

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### BIOCHAR TODAY - Blog Contributions ↗

- Designing biochar from complex wastes: What research is teaching us (2026)
- Advancing biochar research in material science (2025)
- Breathing new life into saline biomass: My story with biochar (2025)

### Invitations & Science Communication

- 1st International Symposium on Sustainable Waste Management and Resource Recovery, University of Sharjah, 2024
- Panel discussion member, American Chemical Society (ACS) Student Symposium, 2024
- Judge, STEAM Fair event, Wesgreen International School, Sharjah, 2022

## Awards

- Green Food Innovation Award, Gulfood Green Terra Award, Dubai, 2024
- Prime Minister's Laptop Award, NUST, 2018
- Green Project Award, UET Peshawar, 2016
- Best Student Award, Islamia English Medium School, Swabi, 2009

## Scholarships - Total Stipend Worth ≈ 382,680 AED

- Full-time PhD scholarship, American University of Sharjah (2021 - 2025)
- Merit-based Master's scholarship, NUST (2017 - 2020)
- Fauji Foundation academic scholarship (2008 - 2016)

## PROJECT AWARDS & INDUSTRIAL LINKS

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### Projects - Funding Value ≈ 1,712,450 AED

- ATA Bayer Liquor Organics Study
- Production of acidic biochar for soil amendment (Emirates Global Aluminium sponsorship)
- Providing equitable, quality and sustained school education and vocational training to vulnerable communities

### Supervision

- Co-supervision of 2 MSc projects, American University of Sharjah
- Supervision/co-supervision of 6 BSc environmental projects, American University of Sharjah

### Peer Review

- Bioresource Technology Reports (9 manuscripts)
- Sustainable Cities and Society (4 manuscripts)
- Fuel (2 manuscripts)
- Journal of Environmental Management (1 manuscripts)

## COURSES, CERTIFICATES & WORKSHOPS

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### Professional Certificates

- Future 17 Sustainable Development Goals Challenge Program, SYSCO & QS Quacquarelli Symonds, 2024
- Certificate in University Teaching, American University of Sharjah, 2022

### Online Courses

- Global Environmental Management, 2020, Coursera
- Mechanics of Materials I: Fundamentals of Stress & Strain and Axial Loading, 2019, Coursera
- Introduction to Household Water Treatment and Safe Storage, 2019, Coursera
- Planning & Design of Sanitation Systems and Technologies, 2019, Coursera
- Municipal Solid Waste Management in Developing Countries, 2019, Coursera
- AutoCAD, 2019, Virtual University (Digiskills)
- Microsoft Office Specialist, 2017, Microsoft

### Workshops

- Student motivation, Centre for Teaching and Learning (CTL), Amity Univesrity Dubai, 17, March, 2026
- Active learning, Centre for Teaching and Learning (CTL), Amity Univesrity Dubai, 13, March, 2026
- Assessment and feedback, Centre for Teaching and Learning (CTL), Amity Univesrity Dubai, 12, March, 2026
- Lesson Planning, Centre for Teaching and Learning (CTL), Amity Univesrity Dubai, 11, March, 2026
- Academic advising + QF Emirates, Centre for Teaching and Learning (CTL), Amity Univesrity Dubai, 10, March, 2026
- NMR Crystallography, New York University Abu Dhabi, 1, May, 2025
- Training and developing employees, Directorate of Worker Education Islamabad 22-24, Sep. 2020
- Hydro Power, National University of Science and Technology, 24-26, Sep. 2018

## TECHNICAL SKILLS

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### Experimental & Analytical

- Pyrolysis reactors (batch tube/auger reactor), biochar production & biogas systems
- Proximate, ultimate analysis, TGA, furnaces, pH meters, ICP-AES, AAS, SEM/EDS, GC-MS, FTIR, BET surface area analysis, zeta potential (Litesizer 500)
- Heavy-metal and organic pollutant analysis; environmental risk and toxicity assessment

### Software & Data

- Microsoft Office (Word, Excel, PowerPoint); MS Project & Primavera P6
- AutoCAD; basic CAD and modeling tools
- Data analysis, visualization, and technical reporting

### Project & Academic Skills

- Project planning and management (scope, cost, schedule, risk)
- Laboratory management, SOP development, safety procedures
- Student training, supervision, and mentoring
- Scientific writing (articles, reports, thesis, blogs)

## CONFERENCES & SEMINARS

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### The International Workshop on Advanced Materials

Ras Al Khaimah, UAE • 02 - 04 February 2026

- Multi-phase energy recovery from Salicornia and date palm biomass via pyrolysis: yield, composition, and energy content

### Materials Today Family of Journals (Elsevier)

Sitges, Spain • 23 - 26 June 2025

- Multi-phase energy recovery from Salicornia and date palm biomass via pyrolysis: yield, composition, and energy content

### GSRC 2025, 10th Graduate Students Research Conference

American University of Sharjah, UAE • 12 April 2025

- Co-pyrolysis of Salicornia and date palm waste: Assessing the impact of temperature on product yield, distribution, and quality

### First UAE - Italy Bilateral Conference: Advanced Materials for the 21st Century

UAE University, Al Ain, UAE • 03 - 04 February 2025

- Advancing sustainable biochar: Risk assessment of PAHs and heavy metals from co-pyrolysis of Salicornia and date palm biomass

### AUS - SIAT Workshop

American University of Sharjah, UAE • 10 - 11 December 2024

- Environmental risk assessment of PAHs and heavy metals in biochar from Salicornia and date palm co-pyrolysis

### AUS Innovation Expo: Future of Research and Technology in the UAE

American University of Sharjah UAE • 06 May 2024

- Environmental risk assessment of heavy metals from co-pyrolysis of Salicornia and date palm (*Phoenix dactylifera*) biochar generated through pyrolysis

### Alumina 2024: 12th International AQW Conference & Exhibition

Dubai, UAE • 21 - 26 April 2024

- Halophytic biomass conversion: Co-pyrolysis of Salicornia and date palm (*Phoenix dactylifera*) for energy products

### Alumina 2024: 12th International AQW Conference & Exhibition

Dubai, UAE • 21 - 26 April 2024

- Biomass pyrolysis: Potential for bauxite residue amendment and mitigation of climate change

### International Conference on Frontiers in Materials Science and Engineering (FMSE 2024)

American University of Sharjah, UAE • 13 - 15 February 2024

- Environmental risk assessment of heavy metals from co-pyrolysis of Salicornia and date palm (*Phoenix dactylifera*) biochar generated through pyrolysis

### 1st Sharjah International Conference on Physics of Advanced Materials (SICPAM)

University of Sharjah, UAE • 25–27 April 2023

- Feasibility study of food waste and sewage sludge to energy through incineration

### Research Day

American University of Sharjah, UAE • 01 March 2023

- Evaluation of pure food waste stream to energy through pyrolysis thermochemical treatment

### 4th International Conference on Energy Conservation and Efficiency - (ICECE 2021)

University of Engineering & Technology Lahore • 15 - 16 March 2021

- Biomass pelletizing: Characterization of cow dung assisted solid recovered bio-fuel from agricultural waste

### International Conference on Mechanical Engineering (ICME)

University of Engineering & Technology Lahore • 04 - 05 September 2019

- Gap analysis of municipal solid waste management system in Peshawar, Pakistan
- Evaluation of stability and rheological behavior of TiO<sub>2</sub> - H<sub>2</sub>O nanofluid

### 3rd Young Researchers National Conference on Water & Environment

USPCAS-W, Mehran University of Engineering & Technology, Jamshoro • 04 - 05 September 2019

- Status of solid waste management in Peshawar, Pakistan
- Assessment of indoor air quality in educational buildings

## JOURNAL PUBLICATIONS

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| <b>Toxicological assessment of hawksbill sea turtle eggs from the Arabian Gulf</b> ↗<br>Marine Pollution Bulletin • Volume 222, Part 1                                                                                        | 2025 |
| <b>Enhanced electrocatalytic hydrogen evolution with CoNiFe(Cr/V) high-entropy alloy electrodes</b> ↗<br>APL Materials • 13, 051114                                                                                           | 2025 |
| <b>Hydrogen production, storage, transportation and utilization for the energy sector: A current status review</b> ↗<br>Journal of Energy Storage • Volume 101, Part A                                                        | 2024 |
| <b>Levels of polycyclic aromatic hydrocarbons and organochlorine pesticides in sea snakes (Elapidae: Hydrophiinae) from Sharjah, United Arab Emirates</b> ↗<br>Marine Pollution Bulletin • Volume 206                         | 2024 |
| <b>Co-pyrolysis of Date Palm Waste and Salicornia bigelovii: Insights for bioenergy development in arid and semi-arid regions</b> ↗<br>ACS Omega • Vol 9/Issue 22                                                             | 2024 |
| <b>Biohydrogen production from waste materials: Mini-review</b> ↗<br>Trends in Ecological and Indoor Environmental Engineering • 1(1):16-23                                                                                   | 2023 |
| <b>Applicability of titanium-based catalysts in the photocatalytic degradation of 2,3,7,8-tetrachlorodibenzofuran</b> ↗<br>Molecules • 28(22), 7488                                                                           | 2023 |
| <b>Socio-economic benefits and policy implications of generating sustainable energy from municipal solid waste in Pakistan</b> ↗<br>Journal of Energy and Climate Change • Volume 5                                           | 2023 |
| <b>Proximate and elemental analysis of solid recovered fuel and comparison with fossil fuels: Physical characteristics and economic benefits</b> ↗<br>Bioresource Technology Reports • Volume 18                              | 2022 |
| <b>Enhancement of the corrosion resistance of mild steel with femtosecond laser nanostructuring and CrCoNi medium-entropy alloy coating</b> ↗<br>Applied Surface Science Advances • Volume 12                                 | 2022 |
| <b>Methane refinement and enhancement using iron oxide, packed-column water scrubbing, and activated charcoal scrubbing techniques: Process optimization</b> ↗<br>Waste and Biomass Valorization • Volume 13, pages 2295–2307 | 2021 |

## GRE (GRADUATE RECORD EXAMINATION)

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Verbal Reasoning: 158

Quantitative Reasoning: 160

## VOLUNTEERING & LEADERSHIP

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### Student Volunteer & Organizer - Accelerating Innovation Through Industry - Academia Partnership

American University of Sharjah, UAE • 25 September 2025

- Supported event organization and logistics for a major innovation-focused academic–industry forum.
- Assisted with coordination of participants, sessions, and event flow to ensure smooth execution.

### Session Chair

10th Graduate Students Research Conference (GSRC 2025), UAE • 12 April 2025

- Chaired conference sessions, facilitated research presentations, and ensured adherence to the program.
- Coordinated with speakers, managed Q&A, and contributed to a professional conference environment.

### Event Organizer - Graduate Student Association

American University of Sharjah • 05 September 2021 - 05 September 2022

### Volunteer - Women Empowerment Initiative

Pakistan U.S. Alumni Network (PUAN), Jamshoro Chapter • 05 September 2019

- Participated in a chapter activity focused on women empowerment and education through graffiti-based awareness.
- Worked with a team of five research assistants on a U.S.-funded project emphasizing gender empowerment and social impact.
- Collaborated with a team of organizers to conduct a community event in Jamshoro, Sindh.

### Member

Provincial Youth Assembly, Swabi Chapter • 04 September 2014 - 01 May 2021

- Served as an ambassador for social work and youth engagement initiatives.
- Part of the National Assembly Speaker's CSR team, contributing to community outreach and development programs.

## REFERENCES

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### Prof. Fatin Samara

Professor of Environmental Sciences

Department of Biology, Chemistry and Environmental Sciences, American University of Sharjah, PO Box 26666, Sharjah, UAE  
+971 6 515 2440  
fsamara@aus.edu

### Prof. Sofian Kanan

Professor of Chemistry & Editor, Scientific Reports

Department of Biology, Chemistry and Environmental Sciences, American University of Sharjah, PO Box 26666, Sharjah, UAE  
+971 6 515 2409  
skanan@aus.edu

### Dr. Muhammad Hassan

Associate Professor

USPCAS-E, National University of Sciences & Technology (NUST)  
+92 343 803 8260  
hassan@uspcase.nust.edu.pk